# Pizza Sales Data Analysis using SQL Queries

**Q1. Retrieve the total number of orders placed.**

```
-- Retrieve the total number of orders placed.
select count(order_id)  as total_orders from orders;
```

- **COUNT(order_id)** → Counts total orders.
- **FROM orders** → Uses the orders table.
- **Result** → Shows the **total number of orders placed**.

Output:

| total_orders |
|---|
| 27778 |

**Q2. Calculate the total revenue generated from pizza sales.**

```
-- Calculate the total revenue generated from pizza sales.
SELECT
    SUM(order_details.quantity * pizzas.price) AS Total_Sales
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id
```

- **quantity × price** → Calculates revenue for each pizza order.
- **SUM()** → Adds all revenues to get **total revenue**.
- **JOIN pizzas** → Gets the **price** of each pizza.

Output:

| Total_Sales |
|---|
| 817689.749999993 |

**Q3. Identify the highest-priced pizza.**

```sql
-- Identify the highest-priced pizza
SELECT pizza_types.name, pizzas.price
FROM pizza_types
JOIN pizzas
ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

| name | price |
|---|---|
| The Greek Pizza | 35.95 |

- **JOIN** → Connects pizza price with pizza name.
- **ORDER BY price DESC** → Sorts pizzas from highest to lowest price.
- **LIMIT 1** → Returns the **most expensive pizza**.

**Q4. Identify the most common pizza size ordered.**

```sql
SELECT pizzas.size,
COUNT(order_details.order_details_id) AS order_count
FROM pizzas
JOIN order_details
ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

| size | order_count |
|---|---|
| L | 18532 |
| M | 15380 |
| S | 14124 |
| XL | 544 |
| XXL | 28 |

- **COUNT()** → Counts how many times each pizza size was ordered.
- **GROUP BY size** → Groups orders by pizza size.
- **ORDER BY order_count DESC** → Highest ordered size first.
- **LIMIT 1** → Returns the **most common pizza size**.

**Q5. List the top 5 most ordered pizza types along with their quantities.**

```sql
SELECT pizza_types.name,
SUM(order_details.quantity) AS total_quantity
FROM order_details
JOIN pizzas
ON order_details.pizza_id = pizzas.pizza_id
JOIN pizza_types
ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.name
ORDER BY total_quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

| name | total_quantity |
|---|---|
| The Classic Deluxe Pizza | 2454 |
| The Barbecue Chicken Pizza | 2433 |
| The Hawaiian Pizza | 2423 |
| The Pepperoni Pizza | 2417 |
| The Thai Chicken Pizza | 2370 |

- **SUM(quantity)** → Calculates total pizzas ordered for each type.
- **JOIN tables** → Combines pizza name with order data.
- **GROUP BY name** → Groups orders by pizza type.
- **ORDER BY total_quantity DESC** → Shows most ordered pizzas first.
- **LIMIT 5** → Returns **top 5 most ordered pizza types**.

**Q6. Join the necessary tables to find the total quantity of each pizza category ordered.**

```sql
SELECT pizza_types.category,
SUM(order_details.quantity) AS total_quantity
FROM order_details
JOIN pizzas
ON order_details.pizza_id = pizzas.pizza_id
JOIN pizza_types
ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.category
ORDER BY total_quantity DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| category | total_quantity |
|---|---|
| Classic | 14890 |
| Supreme | 11979 |
| Veggie | 11645 |
| Chicken | 11046 |

- **JOIN tables** → Combines order data with pizza category.
- **SUM(quantity)** → Calculates total pizzas ordered in each category.
- **GROUP BY category** → Groups results by pizza category.

**Q7. Determine the distribution of orders by hour of the day.**

```
SELECT HOUR(time) AS order_hour,
COUNT(order_id) AS total_orders
FROM orders
GROUP BY order_hour
ORDER BY order_hour;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| order_hour | total_orders |
|---|---|
| 9 | 1 |
| 10 | 8 |
| 11 | 1584 |
| 12 | 3257 |
| 13 | 3159 |
| 14 | 2017 |
| 15 | 1944 |
| 16 | 2467 |
| 17 | 3067 |
| 18 | 3110 |
| 19 | 2618 |
| 20 | 2119 |
| 21 | 1542 |
| 22 | 854 |
| 23 | 31 |

- **HOUR(time)** → Extracts the hour from order time.
- **COUNT(order_id)** → Counts orders in each hour.
- **GROUP BY order_hour** → Groups orders by hour.

**Q8. Join relevant tables to find the category-wise distribution of pizzas.**

```
SELECT pizza_types.category,
COUNT(pizzas.pizza_id) AS total_pizzas
FROM pizzas
JOIN pizza_types
ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.category
ORDER BY total_pizzas DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| category | total_pizzas |
|---|---|
| Veggie | 27 |
| Classic | 26 |
| Supreme | 25 |
| Chicken | 18 |

- **JOIN tables** → Links pizzas with their categories.
- **COUNT(pizza_id)** → Counts pizzas in each category.
- **GROUP BY category** → Groups results by pizza category.

**Q9. Group the orders by date and calculate the average number of pizzas ordered per day.**

```sql
SELECT AVG(total_pizzas_per_day) AS avg_pizzas_per_day
FROM (
    SELECT orders.date,
           SUM(order_details.quantity) AS total_pizzas_per_day
    FROM orders
    JOIN order_details
    ON orders.order_id = order_details.order_id
    GROUP BY orders.date
) AS daily_orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| avg_pizzas_per_day |
|---|
| 179.5894 |

- **SUM(quantity)** → Calculates total pizzas ordered each day.
- **GROUP BY date** → Groups orders by date.
- **AVG()** → Finds the **average pizzas ordered per day**.

**Q10. Determine the top 3 most ordered pizza types based on revenue.**

```sql
SELECT pizza_types.name,
SUM(order_details.quantity * pizzas.price) AS revenue
FROM order_details
JOIN pizzas
ON order_details.pizza_id = pizzas.pizza_id
JOIN pizza_types
ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

| name | revenue |
|---|---|
| The Thai Chicken Pizza | 43413.5 |
| The Barbecue Chicken Pizza | 42792.75 |
| The California Chicken Pizza | 41396.75 |

- **quantity × price** → Calculates revenue for each pizza type.
- **SUM()** → Total revenue generated by each pizza.
- **GROUP BY name** → Groups revenue by pizza type.
- **ORDER BY revenue DESC** → Highest revenue pizzas first.